

Mrinal Mohit

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TECHNICAL SKILLS

Languages: Java, Dart, C++, Python, SQL, JavaScript

Backend & Frameworks: Spring Boot, React, Flask, Flutter

Databases & Cloud: PostgreSQL, MongoDB, AWS, Docker, RabbitMQ

Developer Tools: GitHub, Maven

Hardware/IoT: ESP32, GPS Module, Sensors, Arduino, Raspberry Pi 5

Systems Programming: Linux System Programming, Multithreading, Memory Management

PROJECTS

Web & API Automated Validation Framework (GitHub)

Tech Stack: Java – Burp Suite Extender API – OWASP ZAP – Docker – JWT

- Developed a **Java**-based security testing framework using **Burp Suite Extender API** and **OWASP ZAP** for automated detection of **OWASP Top 10** vulnerabilities across **Web and REST APIs**.
- Implemented exploit modules for **SQLi, XSS, SSRF, CSRF, host header injection**, parameter pollution, prototype pollution, and insecure deserialization.
- Developed proof-of-concept exploits for **JWT tampering** (weak key brute force, alg=none abuse), **session fixation**, and **Broken Access Control via IDOR and privilege-escalation fuzzing**.

Self-Hosted Cloud Infrastructure Platform (PoW)

Tech Stack: Raspberry Pi 5 – Debian Linux – Docker – OMV – Nextcloud – Tailscale – Nginx – GitHub Actions

- Engineered a modular self-hosted cloud platform on a **Raspberry Pi 5**, deploying containerized services including **Immich** and **Nextcloud** for private photo backup, file synchronization, and centralized storage management.
- Designed Docker-based infrastructure with persistent volumes, custom bridge networks, restart policies, and reverse proxy routing using **Nginx**, improving service isolation and simplifying multi-service management.
- Configured secure remote connectivity through a **Tailscale** mesh VPN, eliminating public port exposure while enabling encrypted access across multiple devices.
- Built a self-hosted **CI/CD pipeline** using **GitHub Actions** to automate Docker image builds, testing, and deployments, reducing manual deployment effort by approximately **90%**.

GPS-Based Dynamic Toll Collection System (GitHub)

Tech Stack: Spring Boot – Embedded C – PostgreSQL – Redis – RabbitMQ – Docker – Kubernetes

- Designed and implemented an **Embedded C** firmware for **ESP32-WROOM-32** with **NEO-6M GPS** and **SIM800L GSM** modules; implemented an **MQTT** client with QoS 1 for reliable telemetry transmission, achieving **3 m GPS accuracy** and **99%** message delivery rate
- Built a **Spring Boot application** to process real-time GPS data from **ESP32** devices; implemented **PostGIS geofencing** and dynamic pricing with traffic/weather-based multipliers.
- Implemented **JWT authentication**, **Redis**-based rate limiting, and fraud detection, achieving **95% accuracy**; deployed event-driven architecture using **RabbitMQ** for async processing.

EXPERIENCE

Software Developer Intern

Mar 2025 – Apr 2025

Know Your Trips, Dublin

- Collaborated in a **team of 5 developers**, conducted **15+ code reviews**, and **mentored** team members to deliver **scalable mobile features**, improving sprint velocity by **15%**.
- Integrated **JWT authentication** and **Firebase Cloud Messaging**, strengthening **system reliability** by **30%**.
- **Automated** deployments through **CI/CD**, reducing release cycles by **20%**.
- Contributed to **QA testing** and debugging, enhancing app stability by **25%**.

Tech Stack: Flutter, Dart, GetX, MongoDB, AWS, and Firebase.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (*Credential*)
- Oracle Certified Java Professional (Java SE 17) (*Credential*)
- IBM Introduction to Cloud (*Credential*)
- Certified Ethical Hacker (CEH v13) — Ongoing

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

CGPA: 8.24/10

Expected May 2027

LEADERSHIP AND IMPACT

Technical Head — Mozilla Firefox Club

Feb 2025 — Jul 2025

Vellore Institute of Technology

- Led 20+ member tech division, mentoring peers in open-source, web, and automation development.
- Conducted coding bootcamps and hackathons impacting **150+ students**.
- Enhanced cross-team collaboration and project execution using agile practices.

Co-CURRICULAR ACTIVITIES

- Max. rated 1207 on Codeforces.[\(Link\)](#)
- Participated in hackathons focusing on fintech automation and cloud-based scalability.